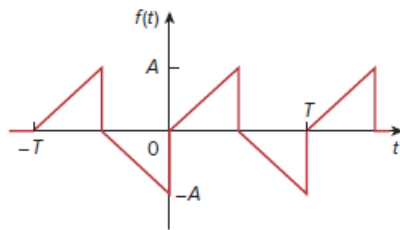


## Problem

Determine the Fourier series of the function in Fig. 17.12(a). Take  $A = 8$  and  $T = 2\pi$ .

**Figure 17.12(a):**



## Step-by-step solution

## Step 1 of 5

Refer to Figure 17.12 in the textbook.

The function  $f(t)$  is expressed as

$$f(t) = \begin{cases} \frac{A}{\pi}t, & 0 < t < \frac{T}{2} \\ -\frac{A}{\pi}t + A, & \frac{T}{2} < t < T \end{cases}$$

Substitute 8 for  $A$ ,  $2\pi$  for  $T$  in the above equation..

$$f(t) = \begin{cases} \frac{8}{\pi}t, & 0 < t < \pi \\ -\frac{8}{\pi}t + 8, & \pi < t < 2\pi \end{cases}$$

Frequency of  $f(t)$  is,

$$\begin{aligned} \omega_0 &= \frac{2\pi}{T} \\ &= \frac{2\pi}{2\pi} \\ &= 1 \end{aligned}$$

## Step 2 of 5

Calculation of  $a_0$ .

$$\begin{aligned}
 a_0 &= \frac{1}{T} \int_0^T f(t) dt \\
 &= \frac{1}{2\pi} \left[ \int_0^{\pi} f(t) dt + \int_{\pi}^{2\pi} f(t) dt \right] \\
 &= \frac{1}{2\pi} \left[ \int_0^{\pi} \left( \frac{8}{\pi} t \right) dt + \int_{\pi}^{2\pi} \left( -\frac{8}{\pi} t + 8 \right) dt \right] \\
 &= \frac{8}{2\pi} \left[ \frac{1}{\pi} \int_0^{\pi} t dt - \frac{1}{\pi} \int_{\pi}^{2\pi} t dt + \int_{\pi}^{2\pi} dt \right] \\
 &= \frac{4}{\pi} \left[ \frac{1}{\pi} \left( \frac{t^2}{2} \right) \Big|_0^{\pi} - \frac{1}{\pi} \left( \frac{t^2}{2} \right) \Big|_{\pi}^{2\pi} + t \Big|_{\pi}^{2\pi} \right] \\
 &= \frac{4}{\pi} \left[ \frac{1}{\pi} \left( \frac{\pi^2}{2} - 0 \right) - \frac{1}{\pi} \left( \frac{4\pi^2}{2} - \frac{\pi^2}{2} \right) + (2\pi - \pi) \right] \\
 &= \frac{4}{\pi} \left[ \frac{\pi}{2} - \frac{3\pi}{2} + \pi \right] \\
 &= 0
 \end{aligned}$$

**Step 3** of 5

Calculation of  $a_n$ .

$$\begin{aligned}
 a_n &= \frac{2}{T} \int_0^T f(t) \cos n\omega_0 t dt \\
 &= \frac{1}{\pi} \left[ \int_0^{\pi} f(t) \cos nt dt + \int_{\pi}^{2\pi} f(t) \cos nt dt \right] \\
 &= \frac{1}{\pi} \left[ \int_0^{\pi} \left( \frac{8}{\pi} t \right) \cos nt dt + \int_{\pi}^{2\pi} \left( -\frac{8}{\pi} t + 8 \right) \cos nt dt \right] \\
 &= \frac{8}{\pi} \left[ \int_0^{\pi} \left( \frac{1}{\pi} t \right) \cos nt dt + \int_{\pi}^{2\pi} \left( -\frac{1}{\pi} t + 1 \right) \cos nt dt \right] \\
 &= \frac{8}{\pi} \left[ \frac{1}{\pi} \left( t \frac{\sin nt}{n} \Big|_0^{\pi} - \frac{1}{n} \int_0^{\pi} \sin nt \right) - \frac{1}{\pi} \left( t \frac{\sin nt}{n} \Big|_{\pi}^{2\pi} - \frac{1}{n} \int_{\pi}^{2\pi} \sin nt \right) + \frac{\sin nt}{n} \Big|_{\pi}^{2\pi} \right] \\
 &= \frac{8}{\pi} \left[ \frac{1}{\pi} \left( \frac{1}{n} (\pi \sin n\pi - 0 \sin 0\pi) + \frac{1}{n} \frac{\cos nt}{n} \Big|_0^{\pi} \right) \right. \\
 &\quad \left. - \frac{1}{\pi} \left( \frac{1}{n} (2\pi \sin 2n\pi - \pi \sin n\pi) + \frac{1}{n} \frac{\cos nt}{n} \Big|_{\pi}^{2\pi} \right) + \frac{1}{n} (\sin 2n\pi - \sin n\pi) \right] \\
 &= \frac{8}{\pi} \left[ \frac{1}{\pi} \left( \frac{1}{n} (0 - 0) + \frac{1}{n^2} ((-1)^n - 1) \right) \right. \\
 &\quad \left. - \frac{1}{\pi} \left( \frac{1}{n} (0 - 0) + \frac{1}{n^2} (1 - (-1)^n) \right) + \frac{1}{n} (0) \right] \\
 &= \frac{16}{\pi^2 n^2} ((-1)^n - 1)
 \end{aligned}$$

**Step 4** of 5

Calculation of  $b_n$ .

$$\begin{aligned}
 b_n &= \frac{2}{T} \int_0^T f(t) \sin n\omega_0 t \, dt \\
 &= \frac{1}{\pi} \left[ \int_0^{\pi} f(t) \sin nt \, dt + \int_{\pi}^{2\pi} f(t) \sin nt \, dt \right] \\
 &= \frac{1}{\pi} \left[ \int_0^{\pi} \left( \frac{8}{\pi} t \right) \sin nt \, dt + \int_{\pi}^{2\pi} \left( -\frac{8}{\pi} t + 8 \right) \sin nt \, dt \right] \\
 &= \frac{8}{\pi} \left[ \frac{1}{\pi} \int_0^{\pi} t \sin nt \, dt - \frac{1}{\pi} \int_{\pi}^{2\pi} t \sin nt \, dt + \int_{\pi}^{2\pi} \sin nt \, dt \right] \\
 &= \frac{8}{\pi} \left[ \frac{1}{\pi} \left( -t \frac{\cos nt}{n} \Big|_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos nt \, dt \right) - \frac{1}{\pi} \left( -t \frac{\cos nt}{n} \Big|_{\pi}^{2\pi} + \frac{1}{n} \int_{\pi}^{2\pi} \cos nt \, dt \right) \right. \\
 &\quad \left. - \frac{\cos nt}{n} \Big|_{\pi}^{2\pi} \right] \\
 &= \frac{8}{\pi} \left[ \frac{1}{\pi} \left( -\frac{\pi}{n} \cos n\pi + \frac{0}{n} \cos n0 + \frac{1}{n} \frac{\sin nt}{n} \Big|_0^{\pi} \right) \right. \\
 &\quad \left. - \frac{1}{\pi} \left( -\frac{2\pi}{n} \cos 2n\pi + \frac{\pi}{n} \cos n\pi + \frac{1}{n} \frac{\sin nt}{n} \Big|_{\pi}^{2\pi} \right) - \frac{\cos nt}{n} \Big|_{\pi}^{2\pi} \right] \\
 &= \frac{8}{\pi} \left[ \frac{1}{\pi} \left( -\frac{\pi}{n} (-1)^n + 0 + \frac{1}{n^2} (0) \right) - \frac{1}{\pi} \left( -\frac{2\pi}{n} + \frac{\pi}{n} (-1)^n + \frac{1}{n^2} (0) \right) \right. \\
 &\quad \left. - \frac{1}{n} (\cos 2n\pi - \cos n\pi) \right] \\
 &= \frac{8}{\pi} \left[ -\frac{1}{n} (-1)^n + \frac{1}{n} \right] \\
 &= \frac{8}{n\pi} [1 - (-1)^n]
 \end{aligned}$$

Step 5 of 5

The function  $f(t)$  is,

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Substitute 0 for  $a_0$ ,  $\frac{16}{\pi^2 n^2}((-1)^n - 1)$  for  $a_n$ ,  $\frac{8}{n\pi}[1 - (-1)^n]$  for  $b_n$  in the above equation.

$$\begin{aligned} f(t) &= 0 + \sum_{n=1}^{\infty} \left[ \left( \frac{16}{n^2 \pi^2} (-1)^n - \frac{16}{n^2 \pi^2} \right) \cos nt + \left( -\frac{8}{n\pi} (-1)^n + \frac{8}{n\pi} \right) \sin nt \right] \\ &= \sum_{n=1}^{\infty} \left[ \left( \frac{16}{n^2 \pi^2} (-1)^n - \frac{16}{n^2 \pi^2} \right) \cos nt + \left( -\frac{8}{n\pi} (-1)^n + \frac{8}{n\pi} \right) \sin nt \right] \\ &= \left[ \left[ \frac{16}{\pi^2} (-1)^1 - \frac{16}{\pi^2} \right] \cos t + \left[ \frac{16}{2^2 \pi^2} (-1)^2 - \frac{16}{2^2 \pi^2} \right] \cos 2t \right. \\ &\quad \left. + \left[ \frac{16}{3^2 \pi^2} (-1)^3 - \frac{16}{3^2 \pi^2} \right] \cos 3t + \dots + \left[ -\frac{8}{\pi} (-1)^1 + \frac{8}{\pi} \right] \sin t \right. \\ &\quad \left. + \left[ -\frac{8}{2\pi} (-1)^2 + \frac{8}{2\pi} \right] \sin 2t + \left[ -\frac{8}{3\pi} (-1)^3 + \frac{8}{3\pi} \right] \sin 3t + \dots \right] \\ &= \left[ -\frac{32}{\pi^2} \cos t + (0) \cos 2t + -\frac{32}{3^2 \pi^2} \cos 3t + \dots \right. \\ &\quad \left. + \frac{16}{\pi} \sin t + (0) \sin 2t + \frac{16}{3\pi} \sin 3t + \dots \right] \\ &= -\frac{32}{\pi^2} \cos t + 0 - \frac{32}{3^2 \pi^2} \cos 3t + \dots + \frac{16}{\pi} \sin t + 0 + \frac{16}{3\pi} \sin 3t + \dots \\ &= -\frac{32}{\pi^2} \cos t - \frac{32}{3^2 \pi^2} \cos 3t + \dots + \frac{16}{\pi} \sin t + \frac{16}{3\pi} \sin 3t + \dots \\ &= \frac{16}{\pi} \sum_{k=1}^{\infty} \frac{-2}{n^2 \pi} \cos nt + \frac{16}{\pi} \sum_{k=1}^{\infty} \frac{1}{n} \sin nt, \quad n = 2k - 1 \\ &= \frac{16}{\pi} \left( \sum_{k=1}^{\infty} \frac{-2}{n^2 \pi} \cos nt + \frac{1}{n} \sin nt \right), \quad n = 2k - 1 \end{aligned}$$

Therefore, the Fourier series of the function  $f(t)$  is

$$\boxed{\frac{16}{\pi} \left( \sum_{k=1}^{\infty} \frac{-2}{n^2 \pi} \cos nt + \frac{1}{n} \sin nt \right), \quad n = 2k - 1.}$$